Soham Chatterjee

Website: sohamch08.github.io
Email: sohamchatterjee999@gmail.com
Institutional Mail: soham.chatterjee@tifr.res.in

Education

Tata Institute of Fundamental Research **2024–Present**
Integrated PhD in Computer Science

Chennai Mathematical Institute **2021–2024**
BSc. in Math and Computer Science

Publications

[1] Soham Chatterjee, Prahladh Harsha, and Mrinal Kumar. *Deterministic list decoding of Reed-Solomon codes*. In *Proc. 58th ACM Symp. on Theory of Computing (STOC)*, pages 2040–2051. 2026. arXiv:2511.05176, eccc:2025/170, doi:10.1145/3798129.3800908.

Dissertation

MSC Project: Exponential Sums and Weil Bounds: A Survey of Elementary Methods and Decoding Applications
Guide: Mrinal Kumar, STCS [PDF]
May, 2026 [Repo]

Research Interests

I am broadly interested in Error Correcting Codes, Pseudorandomness, Algebra and Computation, Computational Complexity Theory. My current work is focused on efficient unique and list decoding algorithms of good codes and their local decodability/correctability.

Talks and Presentations

- **Deterministic List Decoding of Reed-Solomon Codes** with Prahladh Harsha, Mrinal Kumar
  - Talk at CSE Department, IIT Hyderabad
  - Talk at RAAM 2026, IIT Hyderabad
  - Talk at STOC 2026, Salt Lake City, USA
  - Poster at WACT 2026, University of Copenhagen
  - Invited Talk at ACMU Seminar, ISI Kolkata
- **Exponential Sums and Weil Bounds: A Survey of Elementary Methods and Decoding Applications**, TIFR MSc. Project Seminar, 2026 MSc project under Mrinal Kumar

Workshops and Conferences

- RAAM 2026, IIT Hyderabad
- STOC 2026, Salt Lake City, Utah, USA
- WACT 2026, University of Copenhagen
- Complexity Update Meeting, 2026, Institute of Mathematical Sciences, Chennai
- Workshop on High Dimensional Expanders, 2025, LMSI
- FSTTCS 2025 Conference, BITS Goa
- HDX and Codes Workshop, 2025, ICTS Bangalore
- FSTTCS 2024 Conference, IIT Gandhinagar
- Quantum Computing Semester, 2024, Chennai Mathematical Institute

Paper Review

- Reviewer for FSTTCS, 2026.
- Reviewer for ISAAC, 2026.
- Sub-reviewer for STACS, 2026.

Relevent Coursework

TIFR

- Advance Coding Theory
- Algebra, Number Theory & Computation
- Combinatorial Optimization
- Algorithms
- Mathematical Foundations in Computer Science
- Pseudorandomness
- Complexity Theory
- Algorithmic Game Theory
- Probability Theory

CMI

Computer Science Courses:

- Expander Graphs & Its Applications
- Algorithmic Coding Theory
- Quantum Algorithmic Thinking
- Parallel Algorithms
- Theory of Computation
- Complexity Theory
- Algebra & Computation
- Quantum Information Theory
- Design & Analysis of Algorithms
- Discrete Mathematics

Math Courses:

- Commutative Algebra
- Rings and Field Theory (Algebra III)
- Group Theory (Algebra II)
- Linear Algebra (Algebra I)
- Topology
- Metric Space (Analysis III)
- Analysis over Euclidean Space (Analysis II)
- Real Analysis (Analysis I)

Miscellaneous Experience

- Teaching Assistant:
  - Algebra, Number Theory & Computation, 2026 course taken by Mrinal Kumar in TIFR.
  - Algorithms, 2025 course taken by T. Kavitha in TIFR.
- Assisted in the selection process for **Vigyan Vidushi**, TIFR.
- I have volunteered for various outreach activities at TIFR such as Frontiers of Scince and National Science Day between November 2024 to February 2026 to inspire students to pursue pure science as a vocation and showcases recent scientific advances to the general public.